ASSAM SCIENCE AND TECHNOLOGY UNIVERSITY
Guwahati
Course Structure and Syllabus

(From Academic Session 2024-25 onwards)

B. TECH
ELECTRONICS AND TELECOMMUNICATION ENGINEERING

4th SEMESTER

ASSAM SCIENCE AND TECHNOLOGY UNIVERSITY

Course Structure

(From Academic Session 2024-25 onwards)

B. Tech 4th Semester: Electronics and Telecommunication Engineering

Semester IV

Sl. No.	Course Code	Course Type	Course Title	Hours per Week			Credit	Marks	
				L	T	P	C	CE	ESE
Theory									
1	ET241401	PCC	Analog Circuits	3	0	0	3	30	70
2	ET241402	PCC	Analog and Digital Communication	3	0	0	3	30	70
3	ET241403	PCC	Microprocessors	3	0	0	3	30	70
4	ET241404	PCC	Electromagnetic Waves	3	0	0	3	30	70
5	ET241405	PCC	Advanced Programming	2	0	4	4	30	70
6	HS241406	HSMC	Finance and Accounting	3	0	0	3	30	70
7	AU241407	AU	Environmental Science	3	0	0	0 (PP/NP)	100	-
Practical									
8	ET241411	PCC	Analog Circuits Lab	0	0	2	1	15	35
9	ET241412	PCC	Analog and Digital Communication Lab	0	0	2	1	15	35
10	ET241413	PCC	Microprocessors Lab	0	0	2	1	15	35
11	PR241415	PR	Micro Project (Skill Based)	0	0	4	2	15	35
TOTAL				**20**	**0**	**14**	**24**	**340**	**560**
Total Contact Hours per week: 34									
Total Credit: 24									

Course Code	Course Title	Hours per week L-T-P	Credit
ET241401	**Analog Circuits**	**3-0-0**	**3**

Course Outcomes: At the end of the course the students will be able to

- **CO1**: Analyse single stage amplifier circuits.
- **CO2**: Analyse multi stage amplifier circuits.
- **CO3**: Classify power amplifier circuits.
- **CO4**: Differentiate feedback topologies.
- **CO5**: Design sinusoidal and non-sinusoidal oscillators.

MODULE 1:

Amplifier Models:

Voltage amplifier, current amplifier, trans-conductance amplifier and trans-resistance amplifier.

Single Stage Amplifier:

Small signal analysis, low frequency transistor models, estimation of voltage gain, input resistance, output resistance, design procedure for particular specifications, High frequency transistor models, frequency response of single stage amplifier (common source, common emitter, common gate, common base, source follower, emitter follower).

Current mirror:

Basic topology and its variants, V- I characteristics, output resistance and minimum sustainable voltage (VON), maximum usable load. **(12 Lectures)**

MODULE 2:

Multistage Amplifiers:

Classification, Distortion, Low and high Frequency responses and Bode plots, Cascode amplifiers (CE-CB), effect of coupling and bypass capacitors, RC coupled amplifier and its low frequency response. **(12 Lectures)**

MODULE 3:

Power Amplifier:

Various classes of operation (Class A, B, AB and C), their power efficiency and linearity issues. **(7 Lectures)**

MODULE 4:

Feedback Topologies

Voltage series, current series, voltage shunt, current shunt, effect of feedback on gain, bandwidth etc., calculation with practical circuits, concept of stability, gain margin and phase margin. **(7 Lectures)**

MODULE 5:

Oscillators

Barkhausen criterion, RC oscillators (phase shift, Wienbridge), LC oscillators (Hartley, Colpitt, Clapp), crystal oscillator, non-sinusoidal oscillators. **(7 Lectures)**

Text books/Reference books

1. Microelectronic Circuits: Theory and Applications, A. S. Sedra and K. C. Smith, 7th edition. Oxford, 2017.
2. Electronic Devices and Circuit Theory, R. L. Boylestad and L. Nashelsky, 11th edition. Pearson, 2013.
3. Electronic Circuits: Analysis and Design, D. A. Neamen, 3rd edition. Tata McGraw-Hill, 2008

Course Code	Course Title	Hours per week L-T-P	Credit
ET241402	**Analog and Digital Communication**	**3-0-0**	**3**

Course Outcomes: At the end of the course the students will be able to

- **CO1**: Analyze various analog modulation techniques and their behaviour in presence of noise.
- **CO2**: Apply various pulse modulation techniques and method for detecting pulse signal.
- **CO3**: Apply various digital modulation techniques and concepts pertaining to detection of signal in AWGN channel.
- **CO4**: Apply basic concepts of information theory.

MODULE 1:

Review:

Review of signals and systems, frequency domain representation of signal

Analog modulation techniques:

Principles of amplitude modulation systems- DSB (DSB-LC and DSB-SC), SSB and VSB modulations, principles of angle modulation systems- FM and PM, Spectral characteristics of angle modulated signals. **(10 Lectures)**

MODULE 2:

Noise in analog modulation:

Sources and characteristics of different noise, concept of white Gaussian Noise, noise calculation-Noise Temperature, Noise Bandwidth and Noise Figure, noise in amplitude modulation systems, noise in frequency modulation systems, Pre-emphasis and De-emphasis, threshold effect in frequency modulation. **(8 Lectures)**

MODULE 3:

Pulse modulation techniques:

Review of sampling process, pulse amplitude and Pulse Code Modulation (PCM), noise considerations in PCM, Differential Pulse Code Modulation, Delta Modulation, Time Division Multiplexing and Frequency Division Multiplexing. **(5 Lectures)**

MODULE 4:

Baseband pulse transmission:

Introduction to Matched Filter receiver, Inter Symbol Interference, Nyquist criterion. **(5 Lectures)**

MODULE 5:

State-space analysis:

Geometric representation of signal, Gram-Schmidt orthogonalization procedure, optimum detection in AWGN channel- maximum likelihood detection, Correlation Receiver, , error probability for the AWGN channel. **(5 Lectures)**

MODULE 6:

Passband digital modulation techniques:

Coherent Amplitude Shift Keying, Phase Shift Keying and Frequency Shift Keying; M-ary Phase Shift Keying; introduction to Quadrature Amplitude Modulation, Continuous Phase Modulation and Minimum Shift Keying. **(7 Lectures)**

MODULE 7:

Information theory:

Discrete source models – memory-less and stationary, uncertainty, information and entropy, Source Coding Theorem, Huffman coding, discrete memoryless channel, conditional entropy, Mutual Information, channel capacity, Channel Coding Theorem. **(5 Lectures)**

Text books/Reference books

1. Communications Systems, Simon Haykin, John Wiley and Sons, 4th Edition, 2001
2. Modern Digital and Analog Communications Systems, B.P.Lathai, Oxford University Press, 3rd Edition, 1998
3. Communication Systems Engineering, John G. Proakis and Masoud Salehi, Pearson Education, 2nd Edition, 2002
4. Principles of Communication Systems, Herbert Taub and Donald L Schilling, Tata McGraw Hill, 3rd Edition, 2012
5. Digital Communications, John G. Proakis and Masoud Salehi, McGraw Hill, 4th Edition, 2008
6. Fundamentals of Digital Communication, Upamanyu Madhow, Cambridge University Press, 2008

Course Code	Course Title	Hours per week L-T-P	Credit
ET241403	**Microprocessors**	**3-0-0**	**3**

Course Outcomes: At the end of the course the students will be able to

- **CO1:** Explain the architecture of microprocessors and some advanced processors.
- **CO2:** Use assembly language for programming of 8085 Microprocessor.
- **CO3:** Analyze the interrupt structure of 8085 and explain the concepts of vectored interrupts and direct memory access (DMA) techniques.
- **CO4:** Explain the concepts of memory and I/O interfacing with 8085 processor with Programmable devices.
- **CO5:** Describe the features of advanced microprocessors.

MODULE 1: Microprocessor Architecture (5 Lectures)

Microprocessor Architecture and its Operations, Fetching, decoding and execution of an Instruction, 8085 architectures, Memory classification and interfacing, Memory –map, Address decoding; Pin diagram of 8085, interfacing of I/O devices, peripheral I/O and memory mapped I/O.

MODULE 2: Programming of 8085 (20 Lectures)

Instruction Set of 8085:

8085 Programming Model, Instruction Classification, Instruction and Data Format, Addressing Modes. Data Transfer Operations, Arithmetic Operations, Logic Operations, Branch Operations. Instruction cycle, Machine cycle and T state, Bus Timing diagram.

Programming Techniques

Looping, Counting and Indexing, Additional Data Transfer and 16-bit Arithmetic Instructions, Arithmetic Operation related to Memory, Logic Operations: Rotate, Compare, Counters and Time Delays with few examples.

Stacks and Subroutines:

Stack, Subroutine, Restart, conditional call, and return instructions; BCD Addition, BCD Subtraction, Introduction to advanced instructions and applications, multiplication, subtraction with carry.

MODULE 3: Interrupts & Interfacing Data Converters (5 Lectures)

8085 Interrupt, 8085 Vectored Interrupts, Direct Memory Access, Introduction to Digital-to-Analog and Analog-to-Digital Converters, Interfacing 8-bit D/A and A/D Converters.

MODULE 4: Programmable Peripheral Interface Devices (10 Lectures)

8155 Multipurpose Programmable Device (I/O ports & Timer, Interfacing 7-segment-LED), 8255 programmable peripheral interface (block diagram, modes), 8253/8254 Programmable Interval Timer (block diagram, programming 8254), 8259 Programmable Interrupt Controller (block diagram, interrupt operation and features).

MODULE 5: Advanced Processor (5 Lectures)

Concepts of virtual memory, Cache memory, Advanced coprocessor Architectures-286, 486, Pentium

Text books/Reference books

1. Microprocessor Architecture: Programming and Applications with the 8085/8080A, R. S. Gaonkar, Penram International Publishing, 1996
2. Computer Organization and Design The hardware and software interface, D A Patterson and J H Hennessy, Morgan Kaufman Publishers.
3. Microprocessors Interfacing, Douglas Hall, Tata McGraw Hill, 1991

Course Code	Course Title	Hours per week L-T-P	Credit
ET241404	**Electromagnetic Waves**	**3-0-0**	**3**

Course Outcomes: At the end of the course the students will be able to

- **CO1**: Use vector calculus for the analysis of static electric field, magnetic field and electromagnetic field.
- **CO2**: Analyse transmission lines and estimate voltage and current at any point on transmission line for different load conditions.
- **CO3**: Explain the properties of uniform plane wave.
- **CO4**: Estimate reflection and transmission of waves at media interface.
- **CO5**: Explain the waveguides and their modes of operation.

MODULE 1: Review of Electromagnetic Field (10 Lectures)

Vector Calculus: Gradient, Divergence, Curl, Laplacian and Divergence theorem, Stokes theorem.
Electromagnetic field: Basic quantities of Electromagnetics, Basic laws of Electromagnetics: Gauss's law, Ampere's Circuital law, Faraday's law of Electromagnetic induction. Maxwell's equations, Surface charge and surface current, Boundary conditions at media interface.

MODULE 2: Transmission Lines (10 Lectures)

Introduction, Concept of distributed elements, Equations of voltage and current, Standing waves and impedance transformation, Lossless and low-loss transmission lines, Power transfer on a transmission line, Analysis of transmission line in terms of admittances, Transmission line calculations with the
help of Smith chart, Applications of transmission line, Impedance matching using transmission lines

MODULE 3: Uniform Plane Wave (7 Lectures)

Homogeneous unbound medium, Wave equation for time harmonic fields, Solution of the wave equation, Uniform plane wave, Wave polarization, Wave propagation in conducting medium, Phase velocity of a wave, Power flow and Poynting vector

MODULE 4: Plane Waves at Media Interface (10 Lectures)

Plane wave in arbitrary direction, Plane wave at dielectric interface, Reflection and refraction of waves at dielectric interface, Total internal reflection, Wave polarization at media interface, Brewster angle, Fields and power flow at media interface, Lossy media interface, Reflection from conducting boundary

MODULE 5: Waveguides (8 Lectures)

Parallel plane waveguide: Transverse Electric (TE) mode, transverse Magnetic (TM) mode, Cut-off frequency, Phase velocity and dispersion. Transverse Electromagnetic (TEM) mode, Analysis of waveguide -general approach, Rectangular waveguides.

Text books/Reference books

1. Electromagnetics, John D. Krauss, McGraw Hill
2. Electromagnetic Waves, R.K. Shevgaonkar, Tata McGraw Hill India
3. Electromagnetic waves & Radiating Systems, E.C. Jordan & K.G. Balmain, Prentice Hall, India
4. Engineering Electromagnetics, Narayana Rao, Prentice Hall
5. Electromagnetics, David Cheng, Prentice Hall

Course Code	Course Title	Hours per week L-T-P	Credit
ET241405	**Advanced Programming**	**2-0-4**	**4**

Course Outcomes: At the end of the course the students will be able to

- **CO1:** Develop programs using variable, operators, condition, loop, string, list, tuple, dictionaries.
- **CO2:** Design user-defined functions to develop programs by importing modules, external files, and handling exceptions.
- **CO3:** Apply object-oriented concepts while writing programs using regular expressions.
- **CO4:** Apply multiple execution concepts and develop programs.
- **CO5:** Apply networking concepts to design programs using sockets and describe Web Integration, database programming, CGI programming & GUI programming.

MODULE 1: Introduction (8 Lectures)
History of Python, environment setup (IDEs, interpreters, virtual environments). Basics: Syntax, variables, data types, operators. Control Structures: Conditionals, loops, and flow control.
Data Structures: Strings, lists, tuples, and dictionaries (operations, functions, and methods).

MODULE 2: Functions (10 Lectures)
Defining a function, Calling a function, Types of functions, Function Arguments, Anonymous functions, Global and local variables
Modules: Importing module, Math module, Random module, Packages, Composition
Input-Output: Opening and closing file, Reading and writing files
Exception Handling: Exception Definition, Exception Handling, Except clause, Try, finally clause, User Defined Exceptions.

MODULE 3: OOP and Regular Expressions (10 Lectures)
OOP Core Principles & Implementation: Class and objects, Attributes, Inheritance, Overloading, Overriding, Data hiding
Regular Expressions: Match function, Search function, Matching VS Searching, Modifiers, Patterns.

MODULE 4: Multi-Threading (7 Lectures)
Thread, starting a thread, threading module, Synchronizing threads, Multi-threaded Priority Queue, multiprocessing module.

MODULE 5: Networking and Applications (10 Lectures)
Sockets Programming: TCP/UDP sockets, blocking vs. non-blocking I/O.
Client-Server Architecture: Design patterns, concurrency models (threads, processes, async I/O). Internet Modules: HTTP/HTTPS, REST APIs, working with requests (Python) or similar libraries.
Database Programming: CRUD operations, SQL basics. Web Integration: Modern web frameworks (Flask/Django overview). GUI Programming: Tkinter.

Text books/Reference books

- Natural Computing with Python, Giancarlo Zaccone, BPB, ISBN:9789388511612
- Python: The Complete Reference, Martin C. Brown, Tata McGraw Hill ISBN: 9789387572942 3
- Let Us Python, Yashwant Kanetkar, BPB, ISBN: 978-9391392253 4
- Python Programming: A modular approach, Kumar Naveen, Taneja Sheetal, Pearson, ISBN: 978-9352861293 5
- Learning Python, Mark Lutz and David Ascher, O'Reilly, ISBN: 978-1449355739
- First Python, Paul Barry Head, O'Reilly, ISBN: 978-1449382674 7
- Introduction to Computation and Programming Using Python, John Guttag, MIT Press, ISBN: 978-0262529624 8

Course Code	Course Title	Hours per week L-T-P	Credit
HS241406	**Finance and Accounting**	**3-0-0**	**3**

Prerequisites: High School Education

Course Outcome (CO): After completion of the course, the students will be able to

- **CO1:** Develop a strong foundation in accounting principles, including the classification of accounts, the Double Entry System, the application of the Golden Rules of Debit and Credit, record financial transactions in journals, post them into ledgers, and prepare a Trial Balance.
- **CO2:** Understand the role of subsidiary books in accounting, including different types of cash
- books and their preparation; learn to reconcile differences between the Cash Book and Pass Book balances and prepare a Bank Reconciliation Statement
- **CO3:** Differentiate between capital and revenue expenditure, understand the concepts of bad debts and provisions for doubtful debts and gain the skills to prepare final accounts by incorporating necessary adjustments.
- **CO4:** Explore the concepts of savings and investment, including their benefits and differences, identify various investment avenues in physical and financial assets and understand the distinction between trading and investment to make informed financial decisions.
- **CO5:** Have a comprehensive understanding of financial markets, covering money and capital markets, key financial instruments, IPOs, stock exchanges, and mutual funds, and analyze the structure and functions of financial markets and assess different financial instruments used for investment.
- **CO6:** Acquire knowledge and skills in stock market investment and analysis, applying fundamental and technical analysis to evaluate stocks, learn to interpret stock market charts and patterns, assess stock price movements, and utilize various analytical tools for informed investment decisions.

MODULE 1: Introduction to Accounting

Concept and classification of Accounts, Transaction, Double Entry System of Book Keeping, Golden rules of Debit and Credit, Journal- Definition, Advantages, Procedure of Journalising, Ledger, Advantages, Rules regarding Posting, Balancing of ledger accounts, Trial Balance- Definition, Objectives, Procedure of preparation.

MODULE 2: Subsidiary Books, Cash Book and Bank Reconciliation

Name of Subsidiary Books, Cash Book- Definition, Advantages, Objectives, Types of Cash Book, Preparation of different types of Cash Books, Bank
Reconciliation Statement, Reasons of disagreement between Cash Book with Pass Book balance, Preparation of bank Reconciliation Statement.

MODULE 3: Expenditures, Bad Debts and Final Accounts

Concept of Capital Expenditure and Revenue Expenditure, Bad Debts,
Provision for Bad and Doubtful debts, Final Accounts: Preparation of Trading Account, Profit and Loss account and Balance Sheet with adjustments.

MODULE 4: Introduction to Saving and Investment

Savings: Concept, Benefits and saving alternatives, Disadvantage of Saving – Time Value of Money. Investment: Meaning, Advantage, Investment Avenues – Investment in Physical and financial assets, Difference between Saving and Investment; Trading- Meaning, Difference between Trading and Investment.

MODULE 5: Financial Market and Instruments.

Financial Markets and Instruments: Money Market, Capital Market, Money Market Instruments and features. Capital Markets and Features, New Issue Market and Stock Market; New Issue market-Initial Public Offering , Market Players, Secondary Market :Equity and Debentures, BSE and NSE, Index and its uses; Mutual Funds: Meaning and Types.

MODULE 6: Stock Market Investment and Analysis

Stock Market Investment: Fundamental Analysis- Economic Analysis-Industry Analysis and Technical Analysis - Tools of technical analysis, Understanding various types of charts and patterns-(Line chart, Bar Chart, Candlestick Chart Point and Figure Chart)

Textbooks/Reference Books

1. Dam, B. B.Sarda, R.A. Barman, R. Kalita, B., Theory and Practice of Accountancy- Vol. I, Gayatri Publication, 2019
2. Goel, D.K Goel, R. Goel,S. Accountancy, Avichal publishing company, 2020
3. Kevin, S., Security Analysis and Portfolio Management, Prentice Hall India., 2022
4. Pathak,B., Indian Financial System, Pearson Education, 2024
5. Chandra, P., The Invesment Game: How To win, Tata McGrawHill, 2012

Course Code	Course Title	Hours per week L-T-P	Credit
AU241407	**Environmental Science**	**3-0-0**	**0**

Course Outcome (CO): After completion of the course, the students will be able to

- **CO1:** Relate environment and its associated problems
- **CO2:** Explain basic ethics of living and protection of other organisms
- **CO3:** Applying sustainable development principles

MODULE 1: Introduction
Environment and Ecology, Objectives of ecological study, Aspects of Ecology, Autecology, Synecology; Ecosystem, Structural and functional attributes of an ecosystem, Food chain and foodweb, Energy flow, Bio geochemical cycles

MODULE 2: Land: Use and Abuse
Land use: Impact of land–use on environmental quality, Land degradation, Control of land degradation, Wasteland, Wet lands

MODULE 3: Water Pollution
Introduction: Water quality standards, Water pollution

MODULE 4: Air Pollution
Introduction, Air pollution system, Air pollutants, Air pollution laws

MODULE 5: Noise Pollution
Sources of noise pollution, Effects of noise, Physical effects, Physiological effects

MODULE 6: Control of Pollutions
Control of water pollution and management, Water pollution legislations, Water quality management in surface water; Control of air pollution, source correction method, Pollution control equipment; controls of Noise pollution

Textbooks/Reference Books

1. Dr Suresh Dhameja, Environmental engineering and management, 2010
2. Dr B.S. Chauhan, Environmental studies
3. Henry and Hence, Environmental science and engineering
4. Dr Susmitha Baskar, Environmental studies for undergraduate course
5. Clair Sawyer, Chemistry for environmental engineering and science

Course Code	Course Title	Hours per week L-T-P	Credit
ET241411	**Analog Circuits Lab**	**0-0-2**	**1**

Course Outcomes: At the end of the course the students will be able to

- **CO1**: Design analog circuits using BJT/FETs and evaluate their performance characteristics.
- **CO2**: Simulate and analyze analog circuits that uses ICs for different electronic applications.
- **CO3**: Analyze the results, and prepare a formal laboratory report.

Sl. No.	List of Experiments
1	To study the biasing techniques of single-stage BJT amplifiers (Fixed Bias and Voltage-divider Bias).
2	To study BJT common emitter amplifier using voltage-divider bias with feedback, and to determine the gain-bandwidth product from its frequency response.
3	To realize BJT Darlington Emitter follower with and without bootstrapping, and to determine the gain, input and output impedances.
4	To study the working of complementary symmetry class B push-pull power amplifier, and to calculate the efficiency.
5	To study BJT feedback amplifier circuit and calculate the following parameters with and without feedback. i. Mid band gain ii. Bandwidth and cut-off frequencies iii. Input and output impedance.
6	To design tuned oscillator circuit (Hartley or Colpitts) using BJT, and to determine the frequency of oscillation.
7	To study the MOSFET amplifier, and to plot the frequency response.

These experiments may be conducted in two phases
Phase 1: Simulation by appropriate software like Multisim, Pspice, etc.
Phase 2: Hardware implementation.

Course Code	Course Title	Hours per week L-T-P	Credit
ET241412	**Analog and Digital Communication Lab**	**0-0-2**	**1**

Course Outcomes: At the end of the course the students will be able to

- **CO1:** Experiment with various modulation and demodulation circuits.
- **CO2:** Design and implement digital modulation and demodulation techniques.
- **CO3:** Utilize software tools for simulating different related circuits.
- **CO4:** Analyze the results, and prepare a formal laboratory report.

Sl. No.	List of Experiments
1	To study amplitude modulation & demodulation techniques and also to calculate the modulation index.
2	To study frequency modulation & demodulation techniques.
3	To study the working of balanced modulator & demodulator.
4	To generate SSB using phase discrimination method and detection of SSB signal using Synchronous detector.
5	To study Pulse Amplitude Modulation & Demodulation.
6	To study PCM: Generation and Detection.
7	To study DPCM: Generation and Detection.
8	To study Delta Modulation: Generation and Detection.
9	To study amplitude shift keying: Generation and Detection.
10	To study Frequency shift keying: Generation and Detection.
11	To study Binary Phase Shift Keying: Generation and Detection.

Course Code	Course Title	Hours per week L-T-P	Credit
ET241413	**Microprocessors Lab**	**0-0-2**	**1**

Course Outcomes: At the end of the course the students will be able to

- **CO1:** Apply the fundamentals of assembly level / high level language programming of 8085 Microprocessor to perform arithmetic, logical, data transfer applications and branch/loop instructions.
- **CO2:** Interface a microprocessor with peripherals for various applications.
- **CO3:** Analyze the results, and prepare a formal laboratory report.

Sl. No.	List of Experiments
1	To implement the following data transfer operations using 8085 Microprocessor i) Byte and word data transfer in different addressing Modes ii) Block move (with and without overlap) iii) Block interchange
2	To perform the following operations using 8085 Microprocessor i) addition ii) Subtraction iii) multiplication iv) division
3	To write programs using Branch/ Loop instructions i) to add N nos. in array ii) to find largest and smallest nos. in array iii) to sort N numbers in array (ascending and descending order)
4	To write programs involving string manipulation like i) string transfer ii) string reversing iii) string search
5	To write programs to i) read a character from keyboard ii) display of character/ String on console
6	To write programs for interfacing 8085 with the following modules i) Matrix keyboard ii) Seven segment display iii) Stepper motor iv) Analog-to-Digital Converter (8 bit) v) Light dependent resistor (LDR)

Course Code	Course Title	Hours per week L-T-P	Credit
PR241415	**Micro Project (Skill Based)**	**0-0-4**	**2**

Prerequisites: Knowledge of basic science

Objectives:

- To identify problems based on societal or research needs
- To acquaint with the process of solving the problem in a group
- To apply engineering knowledge to attempt solutions to the problems

Course Outcome (CO): After completion of the course, the students will be able to

- CO1: Analyze engineering problems and their solutions
- CO2: Apply modern engineering tools to attempt solutions to the problems
- CO3: Adapt with computational thinking and automation skills
- CO4: Realize the need of professional communications, team work and self-learning

General Guidelines:

1. Students shall form a group of 3 to 4 students.
2. Each group shall conduct field survey or literature survey in consultation with faculty supervisor/mentor or internal project committee of faculty members to identify the needs of the society.
3. Each group shall formulate the problems on the basis of their survey and submit the implementation plan to the concerned faculty supervisor/mentor or the committee.
4. A log book is to be maintained by each group, wherein the group can record the work progress. The supervisor shall monitor the progress.
5. The faculty supervisor/mentor or the committee may give inputs to the students during the project activities, however, focus shall be on self-learning.
6. Students in a group shall understand problem effectively, propose multiple solutions and select the best solution in consultation with faculty supervisor/mentor or the committee.
7. Students shall convert the best solution into a working model/prototype using various components of their domain areas.
8. Students shall demonstrate and validate the model with proper justification and a report is to be compiled in a standard format/template

Assessment Criteria:

The Micro Project shall be assessed based on the following criteria:

Sl No.	Component	Weightage
1	Proper identification of the problem	20
2	Quality of survey	15
3	Clarity of problem definition	10
4	Presence of innovation	5
5	Effective use of modern engineering tools	10
6	Cost effectiveness	5
7	Societal impact	10
8	Functioning of the working model	15
9	Fulfilment of standard engineering norms	5
10	Clarity in written and oral communication	5
		100

Examinations:

Sl No.	Type of examination/evaluation	Marks (50)
1	Mid-term presentation on Progress	10
2	Demonstration of the project	10
3	End-term presentation	25
4	Viva-voce examination	5

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